ANOVA

One way classification:

1. As a crop researcher, you want to test the effect of three different fertilizer mixtures on crop yield. You can use a one-way ANOVA to find out if there is a difference in crop yields between the three groups.
2. Test whether the level of employee training(beginner, intermediate and advanced) impacts the customer satisfaction.
3. Analyze data from experiments that involve testing the effectiveness of a new drug, comparing the survival rates of different treatments for a disease, or analyzing the efficacy of a new medical device.
4. If the three different teaching methods (lecture, workshop, and online learning) affects the student exam scores. The teaching method is the independent variable with three groups, and the exam score is the dependent variable.

Two way classification:

1. The relationship between teaching methods and study techniques and how they jointly affect student performance. The two-way ANOVA is suitable for this scenario. Here we test three hypotheses:

- **The main effect of factor 1 (teaching method)**: Does the teaching method influence student exam scores?
- **The main effect of factor 2 (study technique)**: Does the study technique affect exam scores?
- **Interaction effect**: Does the effectiveness of the teaching method depend on the study technique used?

2. Fertilizer types 1, 2, and 3 are levels within the categorical variable fertilizer type. Planting densities 1 and 2 are levels within the categorical variable planting density.

   Two-way ANOVA hypotheses in our crop yield experiment, we can test three hypotheses using two-way ANOVA:

Null hypothesis (H0)	Alternate hypothesis (Ha)
There is no difference in average yield for any fertilizer type.	There is a difference in average yield by fertilizer type.
There is no difference in average yield at either planting density.	There is a difference in average yield by planting density.
The effect of one independent variable on average yield does not depend on the effect of the other independent variable (a.k.a. no interaction effect).	There is an interaction effect between planting density and fertilizer type on average yield.